Appendix C Processor UNPREDICTABLE behaviors

This appendix describes the specific Cortex®-R82 processor **UNPREDICTABLE** behaviors that differ from the Arm standard behavior.

For each case, the Arm standard specification is listed under *Specification* and the Cortex®-R82 processor implementation is listed under *Implementation*. For detailed background information on **UNPREDICTABLE** behaviors, see *Arm® Architecture Reference Manual for A-profile architecture*.

C.1 SBZ or SBO fields in instructions

This section describes the specification and implementation of SBZ or SBO fields in instructions.

Specification

Some instructions have (0) or (1) in the instruction decode to indicate *should-be-zero*, SBZ, or *should-be-one*, SBO. Except for specific cases identified in **CONSTRAINED UNPREDICTABLE** behaviors with Load-Exclusive/Store-Exclusive pairs as described in the *Arm® Architecture Reference Manual for A-profile architecture*, if the instruction bit pattern of an instruction is executed with these fields not having the *should be* values, one of the following must occur:

- The instruction is **UNDEFINED**
- The instruction executes as a NOP
- The instruction operates as if the bit had the should-be value
- Any destination registers of the instruction become **UNKNOWN**
- For execution at EL0 or EL1, when EL2 is implemented and enabled for the current Security state and HCR_EL2.TIDCP is 1, the instruction is trapped to EL2 with EC value 0

Implementation

SBZ or SBO fields in instructions are treated as *don't care* conditions in the Cortex®-R82 decoders. Therefore, the instructions execute as if the bit is 1 or 0 if it is an SBO or SBZ value irrespective of the actual bit value.

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